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Resin multilayer substrate and method for manufacturing resin multilayer substrate

Abstract

A metal foil layer includes a metal foil layer upper main surface, a metal foil layer lower main surface with a surface roughness greater than that of the metal foil layer upper main surface, and a metal foil layer side surface that is a surface of the metal foil layer excluding the metal foil layer upper main surface and the metal foil layer lower main surface. The metal foil layer upper main surface is in contact with an upper resin layer. The metal foil layer lower main surface is in contact with a lower resin layer. A portion of a lower main surface of the upper resin layer and a portion of an upper main surface of the lower resin layer are in contact with each other to define a layer interface. In a cross section of the metal foil layer, the metal foil layer side surface includes an outer point and an inner point where the upper resin layer is in contact with the side surface, and the inner point is below the outer point and inward of the outer point.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to Japanese Patent Application No. 2020-124365 filed on Jul. 21, 2020 and is a Continuation Application of PCT Application No. PCT/JP2021/026064 filed on Jul. 12, 2021. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a resin multilayer substrate including a metal foil layer and a resin layer.

2. Description of the Related Art

(2) As an invention related to an existing resin multilayer substrate, for example, a multilayer circuit substrate described in International Publication No. 2011/007659 is known. The multilayer circuit substrate includes an upper resin layer, a lower resin layer, and a metal foil layer. The upper resin layer is stacked on the lower resin layer. The metal foil layer is arranged between the upper resin layer and the lower resin layer. An upper main surface of the metal foil layer is in contact with the upper resin layer. A lower main surface of the metal foil layer is in contact with the lower resin layer. A surface roughness of the lower main surface of the metal foil layer is greater than a surface roughness of the upper main surface of the metal foil layer. Accordingly, the strength of the metal foil layer in close contact with the lower resin layer is stronger than the strength of the metal foil layer in close contact with the upper resin layer. As a result, peeling of the metal foil layer from the lower resin layer is prevented.

SUMMARY OF THE INVENTION

(3) However, in the multilayer circuit substrate described in International Publication No. 2011/007659, the metal foil layer and the upper resin layer may be separated from each other. In this case, the upper resin layer and the lower resin layer may be separated from each other.

(4) Preferred embodiments of the present invention provide resin multilayer substrates in each of which an upper resin layer and a lower resin layer are not easily separated from each other.

(5) The inventor of the present application studied the following method to prevent the upper resin layer and the lower resin layer from being separated from each other. Both the main surfaces of the metal foil layer are roughened and both the main surfaces of the metal foil layer are fixed to the resin multilayer substrate. However, the inventor of the present application discovered that in a multilayer circuit substrate using a metal foil layer as a transmission line, when both the main surfaces of the metal foil layer in contact with a resin substrate layer are roughened, transmission loss is more likely to occur in an electric signal flowing through the metal foil layer than when one main surface of the metal foil layer is roughened.

(6) Therefore, the inventor of the present application has studied a method of making it difficult for the upper resin layer and the lower resin layer to be separated from each other and reducing or preventing an increase in the transmission loss, and has conceived of and developed the following preferred embodiments of the present invention.

(7) A resin multilayer substrate according to a preferred embodiment of the present invention

includes a metal foil layer, an upper resin layer, and a lower resin layer, the metal foil layer including a metal foil layer upper main surface, a metal foil layer lower main surface having a surface roughness greater than a surface roughness of the metal foil layer upper main surface, and a metal foil layer side surface that is a surface of the metal foil layer excluding the metal foil layer upper main surface and the metal foil layer lower main surface, the metal foil layer upper main surface being in contact with the upper resin layer, the metal foil layer lower main surface is in contact with the lower resin layer, a portion of a lower main surface of the upper resin layer and a portion of an upper main surface of the lower resin layer being in contact with each other to define a layer interface, in a cross section of the metal foil layer parallel or substantially parallel to a multilayer body up-down direction, a direction from an inner side of a region surrounded by a surface of the metal foil layer toward an outer side portion of a region surrounded by the surface of the metal foil layer of directions perpendicular or substantially perpendicular to the multilayer body up-down direction is defined as an inside-outside direction, in a cross section of the metal foil layer parallel or substantially parallel to a multilayer body up-down direction, the metal foil layer side surface includes an outer point and an inner point located at a portion where the upper resin layer is in contact with the side surface, and the inner point is located below the outer point and is located inward of the outer point in the inside-outside direction.

(8) Hereinafter, definitions of terms in the present specification will be described. In this specification, an axis or a member extending in a front-back direction does not necessarily indicate only an axis or a member parallel or substantially parallel to the front-back direction. The axis or the member extending in the front-back direction is an axis or a member inclined in a range of about $\pm 45^\circ$ with respect to the front-back direction. Similarly, an axis or a member extending in the up-down direction is an axis or a member inclined in a range of about $\pm 45^\circ$ with respect to the up-down direction. An axis or a member extending in a left-right direction is an axis or a member inclined in a range of about $\pm 45^\circ$ with respect to the left-right direction.

(9) Hereinafter, a first member is a member or the like included in a high-frequency signal transmission line. In the present specification, unless otherwise specified, each portion of the first member is defined as follows. A front portion of the first member is a front half of the first member. A back portion of the first member is a back half of the first member. A left portion of the first member is a left half of the first member. A right portion of the first member is a right half of the first member. An upper portion of the first member is an upper half of the first member. A lower portion of the first member is a lower half of the first member. A front end of the first member is an end in a forward direction of the first member. A back end of the first member is an end in a backward direction of the first member. A left end of the first member is an end in a leftward direction of the first member. A right end of the first member is an end in a rightward direction of the first member. An upper end of the first member is an end in an upward direction of the first member. A lower end of the first member is an end in a downward direction of the first member. A front end portion of the first member is the front end of the first member and the vicinity thereof. A back end portion of the first member is the back end of the first member and the vicinity thereof. A left end portion of the first member is the left end of the first member and the vicinity thereof. A right end portion of the first member is the right end of the first member and the vicinity thereof. An upper end portion of the first member is the upper end of the first member and the vicinity thereof. A lower end portion of the first member is the lower end of the first member and the vicinity thereof.

(10) According to the resin multilayer substrates of preferred embodiments of the present invention, the upper resin layer and the lower resin layer are not easily separated from each other.

(11) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an external perspective view of a high-frequency signal transmission line **10** including a bent portion to connect two circuits.
- (2) FIG. 2 is a side view of the high-frequency signal transmission line **10** including a bent portion to connect two circuits.
- (3) FIG. 3 is an external perspective view of the high-frequency signal transmission line **10** including a bent portion.
- (4) FIG. 4 is an external perspective view of the high-frequency signal transmission line **10** that is not bent.
- (5) FIG. 5 is an exploded perspective view of the high-frequency signal transmission line **10** that is not bent.
- (6) FIG. 6 is a cross-sectional view taken along line A-A of FIG. 4.
- (7) FIG. 7 is a cross-sectional view of the high-frequency signal transmission line **10** including an adhesive layer formed of an adhesive.
- (8) FIG. 8 is a cross-sectional view of a high-frequency signal transmission line **10a** according to a first modification of a preferred embodiment of the present invention taken along line A-A.
- (9) FIG. 9 is a cross-sectional view of a high-frequency signal transmission line **10b** according to a second modification of a preferred embodiment of the present invention taken along line A-A.
- (10) FIG. 10 is a cross-sectional view of a high-frequency signal transmission line **10b2** according to the second modification taken along line A-A.
- (11) FIG. 11 is a cross-sectional view of a high-frequency signal transmission line **10c** according to a third modification of a preferred embodiment of the present invention taken along line A-A.
- (12) FIG. 12 is a cross-sectional view of a high-frequency signal transmission line **10d** according to a fourth modification of a preferred embodiment of the present invention taken along line A-A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) A high-frequency signal transmission line **10** according to a first preferred embodiment of the present invention will be described with reference to the drawings. FIG. 1 is an external perspective view of the high-frequency signal transmission line **10** including a bent portion to connect two circuits. FIG. 2 is a side view of the high-frequency signal transmission line **10** including a bent portion connecting two circuits. FIG. 3 is an external perspective view of the high-frequency signal transmission line **10** including a bent portion. FIG. 4 is an external perspective view of the high-frequency signal transmission line **10** that is not bent. FIG. 5 is an exploded perspective view of the high-frequency signal transmission line **10** that is not bent. FIG. 6 is a cross-sectional view taken along line A-A illustrated in FIG. 4. Note that in FIG. 1 and FIG. 2, the high-frequency signal transmission line **10** is indicated by a dot pattern. In addition, in FIG. 2, components provided at an upper end of a circuit A other than the high-frequency signal transmission line **10** are not shown. Further, in FIG. 3, illustration of a circuit to which the high-frequency signal transmission line **10** is connected is omitted.

(14) In addition, in the present specification, directions are defined as follows. A stacking direction of an element body **15** of the high-frequency signal transmission line **10** is defined as a multilayer body up-down direction. A longitudinal direction of the element body **15** is defined as a multilayer body front-back direction. A lateral direction of the element body **15** is defined as a multilayer body left-right direction. The multilayer body up-down direction, the multilayer body front-back direction, and the multilayer body left-right direction are perpendicular or substantially perpendicular to each other. Note that the definitions of the direction and the stacking direction in the present specification are examples. Therefore, the direction in actual use of the high-frequency signal transmission line **10** does not need to coincide with the direction in the present specification.

(15) The high-frequency signal transmission line **10** (corresponding to a resin multilayer substrate) is used to connect two circuits in an electronic device such as a mobile phone. The high-frequency signal transmission line **10** has flexibility. Therefore, the high-frequency signal transmission line **10** can be used in an electronic device in a bent state in the up-down direction (planar direction). For example, as illustrated in FIG. 1 and FIG. 2, the high-frequency signal transmission line **10** can connect the circuit A and a circuit B in the bent state in the up-down direction. Note that the high-frequency signal transmission line **10** having a bent shape also has flexibility. Therefore, the high-frequency signal transmission line **10** in the bent state can be further bent.

(16) As illustrated in FIG. 2 and FIG. 3, the high-frequency signal transmission line **10** includes non-curved sections **A1** and **A3** and a curved section **A2**. The non-curved sections **A1** and **A3** are sections in which the high-frequency signal transmission line **10** is not bent in the up-down direction. The curved section **A2** is a section in which the high-frequency signal transmission line **10** includes a portion that is bent in the up-down direction. The non-curved section **A1** is adjacent to the curved section **A2**. The non-curved section **A1** is located to the left of the curved section **A2**. The non-curved section **A3** is adjacent to the curved section **A2**. The non-curved section **A3** is located to the right of the curved section **A2**.

(17) In this case, as illustrated in FIG. 2 and FIG. 3, the multilayer body up-down direction and the multilayer body front-back direction differ depending on the position of the high-frequency signal transmission line **10**. In the non-curved section **A1** where the high-frequency signal transmission line **10** is not bent (for example, at a position (1) in FIG. 2), the multilayer body up-down direction and the multilayer body front-back direction coincide with the up-down direction and the front-back direction, respectively. On the other hand, in the curved section **A2** where the high-frequency signal transmission line **10** is bent (for example, at a position (2) in FIG. 2), the multilayer body up-down direction and the multilayer body front-back direction do not coincide with the up-down direction and the front-back direction, respectively.

(18) Next, the structure of the high-frequency signal transmission line **10** will be described with reference to FIG. 4 and FIG. 5. Note that FIG. 4 and FIG. 5 illustrate the state where the high-frequency signal transmission line **10** is not bent. In this case, the high-frequency signal transmission line **10** includes only a non-curved section that is not bent. Therefore, in the high-frequency signal transmission line **10** illustrated in FIG. 4 and FIG. 5, the multilayer body up-down direction and the multilayer body front-back direction coincide with the up-down direction and the front-back direction, respectively. As illustrated in FIG. 4 and FIG. 5, the high-frequency signal transmission line **10** includes a resin layer **16a** (corresponding to an upper resin layer), a resin layer **16b** (corresponding to a lower resin layer), resist layers **17a** and **17b** (corresponding to a protective film), a signal conductor layer **18** (corresponding to a metal foil layer), a first ground conductor layer **20** (corresponding to a metal foil layer), a second ground conductor layer **22** (corresponding to a metal foil layer), outer electrodes **24** and **26**, a plurality of first interlayer connection conductors **v1**, a plurality of second interlayer connection conductors **v2**, and interlayer connection conductors **v11** and **v12**.

(19) Note that in FIG. 5, representative interlayer connection conductors among the plurality of first interlayer connection conductors **v1** and the plurality of second interlayer connection conductors **v2** are denoted by reference numerals.

(20) As illustrated in FIG. 5, each of the resin layers **16a** and **16b** has a rectangular or substantially rectangular shape having long sides extending in the front-back direction when viewed in the up-down direction. Therefore, the length of the resin layers **16a** and **16b** in the front-back direction is longer than the length of the resin layers **16a** and **16b** in the left-right direction. The length of the resin layers **16a** and **16b** in the front-back direction is longer than the length of the resin layers **16a** and **16b** in the up-down direction. The resin layers **16a** and **16b** have flexibility. The resin layers **16a** and **16b** are flexible dielectric sheets. The resin layers **16a** and **16b** are made of thermoplastic resin such as polyimide and liquid crystal polymer. Such resin layers **16a** and **16b** define the

element body **15**. More specifically, the element body **15** has a structure in which the resin layers **16a** and **16b** are stacked in the up-down direction. The resin layers **16a** and **16b** are stacked so as to be arranged in this order from top to bottom.

(21) Thus, the element body **15** has a plate shape. As illustrated in FIG. 5, the element body **15** has a rectangular or substantially rectangular shape having longer sides extending in the front-back direction when viewed in the up-down direction. Therefore, the length of the element body **15** in the front-back direction is longer than the length of the element body **15** in the left-right direction. The length of the element body **15** in the front-back direction is longer than the length of the element body **15** in the up-down direction.

(22) The signal conductor layer **18** is provided on an upper main surface of the resin layer **16b**. Thus, the signal conductor layer **18** is provided in the element body **15**. As illustrated in FIG. 5, the signal conductor layer **18** has a linear shape extending in the front-back direction. The signal conductor layer **18** is arranged at the center in the left-right direction of the upper main surface of the resin layer **16b**. A front end of the signal conductor layer **18** is located at a front end portion of the resin layer **16b**. A back end of the signal conductor layer **18** is located at the back end portion of the resin layer **16b**. In the high-frequency signal transmission line **10**, the signal conductor layer **18** is a metal foil layer. The signal conductor layer **18** is a type of circuit pattern. Therefore, the circuit pattern is defined by the metal foil layer. A high-frequency signal is transmitted to the signal conductor layer **18**.

(23) The first ground conductor layer **20** is provided on an upper main surface of the resin layer **16a**. Thus, the first ground conductor layer **20** is located on the signal conductor layer **18**. In the present specification, the state “the first ground conductor layer **20** is located on the signal conductor layer **18** in the up-down direction” is defined as follows. At least a portion of the first ground conductor layer **20** is located in a region through which the signal conductor layer **18** passes when moving in parallel or substantially parallel in the upward direction. Thus, the first ground conductor layer **20** may be within the region through which the signal conductor layer **18** passes when moving in parallel or substantially parallel in the upward direction, or may project from the region through which the signal conductor layer **18** passes when moving in parallel or substantially parallel. In the present preferred embodiment, the first ground conductor layer **20** projects from the region through which the signal conductor layer **18** moves in parallel or substantially parallel.

(24) As illustrated in FIG. 5, the first ground conductor layer **20** has a rectangular or substantially rectangular shape having long sides extending in the front-back direction when viewed in the up-down direction. The first ground conductor layer **20** has a shape that substantially coincides with the shape of the resin layer **16a** when viewed in the up-down direction. However, the first ground conductor layer **20** is slightly smaller than the resin layer **16a** when viewed in the up-down direction. A ground is connected to the first ground conductor layer **20**.

(25) The second ground conductor layer **22** is provided on a lower main surface of the resin layer **16b**. Thus, the second ground conductor layer **22** is located under the signal conductor layer **18**. As illustrated in FIG. 5, the second ground conductor layer **22** has a rectangular or substantially rectangular shape having longer sides extending in the front-back direction when viewed in the up-down direction. The second ground conductor layer **22** has a shape that substantially coincides with the shape of the resin layer **16b** when viewed in the up-down direction. However, the second ground conductor layer **22** is slightly smaller than the resin layer **16b** when viewed in the up-down direction. A ground is connected to the second ground conductor layer **22**. The first ground conductor layer **20** and the second ground conductor layer **22** described above have a strip line structure.

(26) The outer electrode **24** is provided at the front end portion of the lower main surface of the resin layer **16b**. The outer electrode **24** has a rectangular or substantially rectangular shape when viewed in the up-down direction. The outer electrode **24** overlaps a front end portion of the signal conductor layer **18** when viewed in the up-down direction. The second ground conductor layer **22**

is not provided around the outer electrode **24** so that the outer electrode **24** is insulated from the second ground conductor layer **22**. The outer electrode **26** has a structure symmetrical to the outer electrode **24** in the front-back direction. Therefore, the description of the outer electrode **26** is omitted. The signal conductor layer **18**, the first ground conductor layer **20**, the second ground conductor layer **22**, and the outer electrodes **24** and **26** described above are formed by, for example, etching a copper foil provided on the upper main surface or the lower main surface of each of the resin layers **16a** and **16b**.

(27) The plurality of first interlayer connection conductors **v1** is provided in the element body **15** so as to be located on the left of the signal conductor layer **18**. The plurality of first interlayer connection conductors **v1** is arranged in a row at equal or substantially equal intervals in the front-back direction. The plurality of first interlayer connection conductors **v1** penetrates the resin layers **16a** and **16b** in the up-down direction. Upper ends of the plurality of first interlayer connection conductors **v1** are connected to the first ground conductor layer **20**. Lower ends of the plurality of first interlayer connection conductors **v1** are connected to the second ground conductor layer **22**. Thus, the plurality of first interlayer connection conductors **v1** electrically connects the first ground conductor layer **20** and the second ground conductor layer **22**.

(28) The plurality of second interlayer connection conductors **v2** is provided in the element body **15** so as to be located on the right of the signal conductor layer **18**. The plurality of second interlayer connection conductors **v2** is arranged in a row at equal or substantially equal intervals in the front-back direction. The plurality of second interlayer connection conductors **v2** penetrates the resin layers **16a** and **16b** in the up-down direction. The upper ends of the plurality of second interlayer connection conductors **v2** are connected to the first ground conductor layer **20**. The lower ends of the plurality of second interlayer connection conductors **v2** are connected to the second ground conductor layer **22**. Thus, the plurality of second interlayer connection conductors **v2** electrically connects the first ground conductor layer **20** and the second ground conductor layer **22**.

(29) The interlayer connection conductor **v11** is provided at a front end portion of the element body **15**. The interlayer connection conductor **v11** penetrates the resin layer **16b** in the up-down direction. An intermediate portion of the interlayer connection conductor **v11** is connected to the front end portion of the signal conductor layer **18**. A lower end of the interlayer connection conductor **v11** is connected to the outer electrode **24**. Thus, the interlayer connection conductor **v11** electrically connects the signal conductor layer **18** and the outer electrode **24**. Note that the interlayer connection conductor **v12** has a structure symmetrical with the interlayer connection conductor **v11** in the front-back direction. Therefore, description of the interlayer connection conductor **v12** will be omitted. The interlayer connection conductors **v1**, **v2**, **v11**, and **v12** are through-hole conductors. The through-hole conductors are formed by plating through-holes formed in the resin layers **16a** and **16b**. Note that the interlayer connection conductors **v1**, **v2**, **v11**, and **v12** may be via-hole conductors.

(30) The resist layers **17a** and **17b** have the same rectangular or substantially rectangular shape as the element body **15** when viewed in the up-down direction. The resist layers **17a** and **17b** are not portion of the element body **15**. The material of the resist layers **17a** and **17b** is different from the material of the element body **15**. That is, the resist layers **17a** and **17b** are layers different from the element body **15**. As illustrated in FIG. 5 and FIG. 6, the resist layer **17a** covers the surface of the element body **15**. To be specific, the resist layer **17a** is provided on an upper main surface of the element body **15**. The resist layer **17a** is a protective film and covers the first ground conductor layer **20**. Thus, the resist layer **17a** protects the first ground conductor layer **20**. However, since a lower main surface of the first ground conductor layer **20** is fixed to the element body **15**, the lower main surface is not covered with the resist layer **17a**. The resist layer **17a** like this is a resist, a coverlay, or the like. For example, when the resist layer **17a** is a resist, the upper main surface of the resin layer **16a** is coated with the resist. When the resist layer **17a** is a coverlay, the resin layer **16a** and the resist layer **17a** are bonded to each other by an adhesive (not shown) or the like.

(31) As illustrated in FIG. 5 and FIG. 6, the resist layer **17b** covers the surface of the element body **15**. To be specific, the resist layer **17b** is provided under a lower main surface of the element body **15**. The resist layer **17b** covers the second ground conductor layer **22**. Thus, the resist layer **17b** protects the second ground conductor layer **22**. However, since an upper main surface of the second ground conductor layer **22** is fixed to the element body **15**, the upper main surface is not covered with the resist layer **17b**. Similar to the resist layer **17a**, the resist layer **17b** is a resist, a coverlay, or the like.

(32) Openings **h11** to **h18** are provided in the resist layer **17b**. When the resist layer **17b** is viewed in the downward direction, the opening **h11** overlaps the outer electrode **24**. When the resist layer **17b** is viewed in the downward direction, the opening **h15** overlaps the outer electrode **26**. Thus, the outer electrodes **24** and **26** are exposed to the outside from the high-frequency signal transmission line **10** through the openings **h11** and **h15**, respectively. The opening **h12** is provided to the right of the opening **h11**. The opening **h13** is provided in front of the opening **h11**. The opening **h14** is provided to the left of the opening **h11**. Thus, the second ground conductor layer **22** is exposed to the outside from the high-frequency signal transmission line **10** through the openings **h12** to **h14**. Note that the openings **h16** to **h18** have structures symmetrical to the openings **h12** to **h14** in the front-back direction. Therefore, description of the openings **h16** to **h18** will be omitted.

(33) Openings **h19** and **h20** are provided in the second ground conductor layer **22**. The diameters of the openings **h19** and **h20** are larger than the diameters of the outer electrodes **24** and **26**. Therefore, when viewed in the up-down direction, the outer electrode **24** and the outer electrode **26** are not inscribed in the openings **h19** and **h20**, respectively. Thus, the outer electrodes **24** and **26** and the second ground conductor layer **22** are not short-circuited.

(34) Meanwhile, the high-frequency signal transmission line **10** has a structure that prevents the resin layer **16a** and the resin layer **16b** from being separated from each other. This structure will be described in more detail below.

(35) As illustrated in FIG. 6, the signal conductor layer **18** includes a signal conductor layer upper main surface **S1** (metal foil layer upper main surface), a signal conductor layer lower main surface **S2** (metal foil layer lower main surface), and a signal conductor layer side surface **S3** (metal foil layer side surface). In the present specification, the signal conductor layer upper main surface **S1** is a portion that is visible when the signal conductor layer **18** is viewed in the downward direction. The signal conductor layer lower main surface **S2** is a portion that is visible when the signal conductor layer **18** is viewed in the upward direction. In addition, the signal conductor layer side surface **S3** is a surface of the signal conductor layer **18** excluding the signal conductor layer upper main surface **S1** and the signal conductor layer lower main surface **S2**. Note that the signal conductor layer **18** may have any shape as long as the shape is defined by the signal conductor layer upper main surface **S1**, the signal conductor layer lower main surface **S2**, and the signal conductor layer side surface **S3**. In the present preferred embodiment, the signal conductor layer **18** has a rectangular or substantially rectangular shape defined by one signal conductor layer upper main surface **S1**, one signal conductor layer lower main surface **S2**, and four signal conductor layer side surfaces **S3**.

(36) The signal conductor layer lower main surface **S2** of the signal conductor layer **18** is in contact with the resin layer **16b**. More precisely, the signal conductor layer lower main surface **S2** of the signal conductor layer **18** is in contact with the upper main surface of the resin layer **16b**. The signal conductor layer lower main surface **S2** of the signal conductor layer **18** is roughened. Therefore, it is possible to prevent separation between the signal conductor layer lower main surface **S2** of the signal conductor layer **18** and the resin layer **16b**. In addition, the signal conductor layer upper main surface **S1** of the signal conductor layer **18** is in contact with the resin layer **16a**. More precisely, the signal conductor layer upper main surface **S1** of the signal conductor layer **18** is in contact with the lower main surface of the resin layer **16a**. The upper main surface of the signal conductor layer **18** is not roughened to prevent an increase in transmission loss of electric

signals. Therefore, the resin layer **16a** is not fixed to the signal conductor layer **18**. Thus, the close-contact strength of the signal conductor layer **18** to the resin layer **16b** is stronger than the close-contact strength of the signal conductor layer **18** to the resin layer **16a**.

(37) The signal conductor layer **18** is provided between the resin layer **16a** and the resin layer **16b**. As such, the resin layer **16a** and the resin layer **16b** are not in contact with each other over the entire lower main surface of the resin layer **16a** and the entire upper main surface of the resin layer **16b**. Therefore, a portion of the lower main surface of the resin layer **16a** is in contact with a portion of the upper main surface of the resin layer **16b**. Thus, a layer interface RI between the resin layer **16a** and the resin layer **16b** is formed. The layer interface RI is provided at a portion where the lower main surface of the resin layer **16a** and the upper main surface of the resin layer **16b** are in contact with each other. Therefore, the layer interface RI is not located in a portion where the signal conductor layer **18** is provided.

(38) Here, the shape of the side surface of the signal conductor layer **18** will be described in more detail. Hereinafter, attention is paid to the right side surface of the signal conductor layer **18**. First, in a cross section parallel or substantially parallel to the multilayer body up-down direction (i.e., the cross section of FIG. 6), a direction from an inner side of a region surrounded by the surfaces of the signal conductor layer **18** toward an outer side portion of the region surrounded by the surfaces of the signal conductor layer **18** among directions perpendicular or substantially perpendicular to the multilayer body up-down direction is defined as an inside-outside direction. Specifically, in a cross section parallel or substantially parallel to the multilayer body up-down direction, a point located at the inner side of the region surrounded by the surfaces of the signal conductor layer **18** is defined as a point X. A point located at the outer side portion of the region surrounded by the surfaces of the signal conductor layer **18** is defined as a point Y. The position of the point X in the multilayer body up-down direction is the same as the position of the point Y in the multilayer body up-down direction. At this time, the inside-outside direction is a direction from the point X toward the point Y (see FIG. 6). On the right side surface of the signal conductor layer **18**, the inside-outside direction is the rightward direction.

(39) The side surface of the signal conductor layer **18** has a cross-sectional shape obtained by rotating the V-shape by 90°. Specifically, the side surface of the signal conductor layer **18** includes an inclined surface extending outward and upward from the center of the side surface of the signal conductor layer **18** in the multilayer body up-down direction and an inclined surface extending outward and downward from the center of the side surface of the signal conductor layer **18** in the multilayer body up-down direction. As described above, since the side surface of the signal conductor layer **18** has the inclined surface extending outward and upward from the center of the side surface of the signal conductor layer **18** in the multilayer body up-down direction, the signal conductor layer **18** has a structure described below.

(40) The side surface of the signal conductor layer **18** includes an outer point OP and an inner point IP in a cross section parallel or substantially parallel to the multilayer body up-down direction. The outer point OP is located at a portion where the resin layer **16a** is in contact with the side surface. As illustrated in FIG. 6, the right side surface of the signal conductor layer **18** includes the outer point OP in a cross section perpendicular or substantially perpendicular to the front-back direction. The side surface of the signal conductor layer **18** includes the inner point IP in a cross section parallel or substantially parallel to the multilayer body up-down direction. The inner point IP is located at a portion where the resin layer **16a** is in contact with the side surface. The inner point IP is located below the outer point OP and is located inward of the outer point OP in the inside-outside direction. On the right side surface of the signal conductor layer **18** illustrated in FIG. 6, the inside-outside direction is the rightward direction. Therefore, the right side surface of the signal conductor layer **18** includes the inner point IP located below and to the left of the outer point OP in the cross section perpendicular or substantially perpendicular to the front-back direction.

(41) The outer point OP and the inner point IP are located at portions where the side surface of the

signal conductor layer **18** and the resin layer **16a** are in contact with each other. Therefore, the outer point OP and the inner point IP are not located in a portion where the side surface of the signal conductor layer **18** and the resin layer **16b** are in contact with each other. In other words, the outer point OP and the inner point IP are not located below the layer interface RI. Note that since the left side surface of the signal conductor layer **18** has a bilaterally symmetrical structure with respect to the right side surface of the signal conductor layer **18**, description thereof will be omitted.

(42) According to the high-frequency signal transmission line **10**, the resin layer **16a** is prevented from being peeled off from the resin layer **16b**. More specifically, in a cross section of the signal conductor layer **18** parallel or substantially parallel to the multilayer body up-down direction, the side surface of the signal conductor layer **18** has the outer point OP and the inner point IP located at a portion where the resin layer **16a** is in contact with the side surface of the signal conductor layer **18**. Further, the inner point IP is located below the outer point OP and is located inward of the outer point OP in the inside-outside direction. Therefore, the resin layer **16a** is located under a portion P0 between the outer point OP and the inner point IP on the side surface of the signal conductor layer **18**. As such, even when the resin layer **16a** is about to peel off from the signal conductor layer **18**, the resin layer **16a** is caught by the portion P0. Thus, the resin layer **16a** is prevented from being peeled off from the signal conductor layer **18**. As a result, peeling of the resin layer **16a** from the resin layer **16b** due to peeling of the resin layer **16a** from the signal conductor layer **18** is prevented.

(43) In addition, since only the lower main surface of the signal conductor layer **18** is roughened, it is also possible to prevent or reduce an increase in transmission loss of an electric signal.

(44) Method of Manufacturing High-Frequency Signal Transmission Line **10**

(45) Hereinafter, a non-limiting example of a method of manufacturing the high-frequency signal transmission line **10** will be described with reference to FIG. 6.

(46) First, in a first step, the resin layer **16a** including the metal foil layer attached to the upper main surface and the resin layer **16b** including the metal foil layer attached to the upper main surface and the lower main surface are prepared.

(47) Next, in a second step, the metal foil layer attached to the upper main surface of the resin layer **16a** is etched to form the first ground conductor layer **20**. The signal conductor layer **18**, the second ground conductor layer **22**, and the outer electrodes **24** and **26** are formed by etching the metal foil layers attached to the upper main surface and the lower main surface of the resin layer **16b**. In the etching process, for example, a mask having the same shape as the first ground conductor layer **20** is formed on the metal foil layer attached to the upper main surface of the resin layer **16a**, and etching treatment is applied on the metal foil layer on which the mask is formed. Thus, the first ground conductor layer **20** is formed. Similarly, the metal foil layers attached to the upper main surface and the lower main surface of the resin layer **16b** are etched and the signal conductor layer **18**, the second ground conductor layer **22**, and the outer electrodes **24** and **26** are formed.

(48) Next, in a third step, the signal conductor layer **18** is deformed and processed, thus forming a side surface of the signal conductor layer **18** including the outer point OP located on the outermost side and the inner point IP located below the outer point OP and located on the inner side in the inside-outside direction. The third step is, for example, a wet blasting method.

(49) Next, as a fourth step, the resin layer **16a** is stacked on the signal conductor layer **18** and the resin layer **16b**. To be specific, the resin layer **16a** is hot-pressed from above the signal conductor layer **18** and the resin layer **16b**. The resin layer **16a** is made of thermoplastic resin (for example, liquid crystal polymer). Therefore, the resin layer **16a** softens and flows during hot pressing. The fluidized resin layer **16a** enters under the portion P0 between the outer point OP and the inner point IP. Then, the temperature of the resin layer **16a** that has entered under the portion P0 returns to room temperature and the resin layer **16a** solidifies. At this time, the solidified resin layer **16a** is caught by the portion P0. Therefore, the resin layer **16a** is prevented from being peeled off from the

resin layer **16b**. That is, in the present step, when the outer point OP and the inner point IP are formed such that the fluidized resin layer **16a** enters under the portion **P0** between the outer point OP and the inner point IP, the resin layer **16a** is prevented from being peeled off from the resin layer **16b**. As described above, in the fourth step, by using the resin layer **16a** made of thermoplastic resin, it is possible to cause the resin layer **16a** to easily enter under the portion **P0**. (50) Note that the high-frequency signal transmission line **10** may include an adhesive layer between the resin layer **16a** and the resin layer **16b**. To be specific, as illustrated in FIG. 7, in the high-frequency signal transmission line **10**, instead of the resin layer **16a**, an adhesive layer **19** may be stacked on the signal conductor layer **18** and the resin layer **16b**. The adhesive layer **19** is formed of a material that solidifies at a temperature lower than the melting temperature of the resin layer **16b**. The adhesive layer **19** is, for example, a thermoplastic material such as LCP or fluorine resin having a melting temperature lower than that of the resin layer **16b**. Alternatively, the adhesive layer **19** is made of, for example, a thermosetting material such as an elastomer material or an epoxy resin having high fluidity even at a temperature lower than the melting temperature of the resin layer **16b**. At this time, the resin layer **16a** is stacked on the adhesive layer **19**. In this case, the adhesive layer **19** corresponds to the upper resin layer.

(51) When the resin layer **16a** and the resin layer **16b** are made of the same material, there is a possibility that the resin layer **16b** fixed to the signal conductor layer **18** may also flow and change in shape during hot pressing, as with the resin layer **16a**. When the resin layer **16b** flows and changes in shape, the electrical characteristics of the high-frequency signal transmission line **10** may change. On the other hand, the adhesive layer **19** solidifies at a temperature lower than the melting temperature of the resin layer **16b**. Therefore, by performing hot pressing at a temperature at which the resin layer **16b** does not flow and the adhesive layer **19** flows, it is possible to reduce the possibility that the shape of the resin layer **16b** changes. As a result, when the adhesive layer **19** is used, it is possible to manufacture the high-frequency signal transmission line **10** while reducing the possibility of a change in electrical characteristics as compared with the case where the resin layer **16a** is used.

(52) Next, through-holes are formed by irradiating the resin layers **16a** and **16b** with a laser beam. Then, the interlayer connection conductors **v1**, **v2**, **v11**, and **v12** are formed by forming conductors in the through-holes by plating. Note that the through-hole may be formed by a drill.

(53) Finally, the openings **h11** to **h18** and the resist layers **17a** and **17b** are formed. Through the above steps, the high-frequency signal transmission line **10** is completed.

(54) According to the non-limiting example of a manufacturing method of the high-frequency signal transmission line **10**, it is possible to easily manufacture the high-frequency signal transmission line **10**. More specifically, by stacking the resin layer **16a** on the signal conductor layer **18** and the resin layer **16b** formed in the second step, at least a portion of the resin layer **16b** enters under the side surface of the signal conductor layer **18** where the outer point OP is formed. As a result, the high-frequency signal transmission line **10** can be easily manufactured.

(55) In addition, according to the wet blasting method, the outer point OP and the inner point IP can be easily formed on all the side surfaces of the signal conductor layer **18**. More specifically, the upper main surface of the signal conductor layer **18** is crushed by the wet blasting method. When the upper main surface of the signal conductor layer **18** is crushed, the side surface of the signal conductor layer **18** protrudes in the inside-outside direction, and the outer point OP and the inner point IP can be formed on all the side surfaces of the signal conductor layer **18**.

(56) First Modification

(57) Hereinafter, a high-frequency signal transmission line **10a** according to a first modification of a preferred embodiment of the present invention will be described with reference to the drawings. FIG. 8 is a cross-sectional view of the high-frequency signal transmission line **10a** taken along line A-A. Note that FIG. 4 is referred to for the external perspective view of the high-frequency signal transmission line **10a**.

(58) As illustrated in FIG. 8, the high-frequency signal transmission line **10a** is different from the high-frequency signal transmission line **10** in that the shape of a signal conductor layer **18a** is different from the shape of the signal conductor layer **18**. Note that since other configurations of the high-frequency signal transmission line **10a** are the same as those of the high-frequency signal transmission line **10**, descriptions thereof will be omitted.

(59) The side surface of the signal conductor layer **18a** has a cross-sectional shape that can be defined by one oblique line. To be specific, the side surface of the signal conductor layer **18a** includes an inclined surface extending straight from an outer point **OP1** located at an upper end of the side surface of the signal conductor layer **18** in the multilayer body up-down direction to an inner point **IP1** located below the outer point **OP1** and located inside the outer point **OP1**. In this case, the resin layer **16a** is located below a portion **P1** between the outer point **OP1** and the inner point **IP1** on the side surface of the signal conductor layer **18a**. Therefore, the resin layer **16a** is caught by the portion **P1**. Accordingly, the resin layer **16a** is prevented from being peeled off from the resin layer **16b**.

(60) Second Modification

(61) Hereinafter, a high-frequency signal transmission line **10b** according to a second modification of a preferred embodiment of the present invention will be described with reference to the drawings. FIG. 9 is a cross-sectional view of the high-frequency signal transmission line **10b** taken along line A-A. FIG. 10 is a cross-sectional view of a high-frequency signal transmission line **10b2** taken along line A-A. Note that FIG. 4 is referred to for the external perspective view of the high-frequency signal transmission line **10b**.

(62) As illustrated in FIG. 9, the high-frequency signal transmission line **10b** is different from the high-frequency signal transmission line **10** in that the shape of a signal conductor layer **18b** is different from the shape of the signal conductor layer **18**. Note that since other configurations of a high-frequency signal transmission line **10b** are the same as those of the high-frequency signal transmission line **10**, descriptions thereof will be omitted.

(63) A protrusion protruding outward is provided in the vicinity of the center of the right side surface of the signal conductor layer **18b**. An outer point **OP2** is located at the protrusion. An inner point **IP2** is located at a portion where the right side surface of the signal conductor layer **18b** and the layer interface **RI** are in contact with each other. Also in the present modification, the resin layer **16a** is located below a portion **P2** between the outer point **OP2** and the inner point **IP2**. Therefore, the resin layer **16a** is caught by the protrusion. Accordingly, the resin layer **16a** is prevented from being peeled off from the resin layer **16b**.

(64) Note that the shape of the protrusion may be any shape. In addition, the protrusion does not necessarily extend in a direction parallel or substantially parallel to the multilayer body left-right direction. In the example illustrated in FIG. 9, the protrusion has a rectangular or substantially rectangular shape extending in a multilayer body rightward direction. However, for example, as illustrated in FIG. 10, the protrusion may have a triangular prism shape extending in the multilayer body rightward direction. In this case, as illustrated in FIG. 10, the protrusion has a triangular shape when viewed in the front-back direction. To be specific, the protrusion includes a first surface **SS1** and a second surface **SS2**. The first surface **SS1** is located above the second surface **SS2**. The second surface **SS2** extends in the inside-outside direction when viewed in the front-back direction. When viewed in the front-back direction, the first surface **SS1** extends in the inside-outside direction and in the multilayer body downward direction. When viewed in the front-back direction, the first surface **SS1** and the second surface **SS2** are in contact with each other at the outer point **OP2**. Even in this case, the resin layer **16a** is located below the portion **P2** between the outer point **OP2** and the inner point **IP2**. Therefore, the resin layer **16a** is prevented from being peeled off from the resin layer **16b**.

(65) Third Modification

(66) Hereinafter, the high-frequency signal transmission line **10c** according to a third modification

of a preferred embodiment of the present invention will be described with reference to the drawings. FIG. 11 is a cross-sectional view of the high-frequency signal transmission line 10c taken along line A-A. Note that FIG. 4 is referred to for the external perspective view of the high-frequency signal transmission line 10c.

(67) As illustrated in FIG. 11, the high-frequency signal transmission line 10c is different from the high-frequency signal transmission line 10 in that the shape of a first ground conductor layer 20b is different from the shape of the first ground conductor layer 20 and the shape of a second ground conductor layer 22b is different from the shape of the second ground conductor layer 22. Note that since other configurations of the high-frequency signal transmission line 10c are the same as those of the high-frequency signal transmission line 10, descriptions thereof will be omitted.

(68) The side surface of the first ground conductor layer 20b includes an outer point OP3 located outermost in a portion where the resist layer 17a is in contact with the side surface in the inside-outside direction. In addition, the side surface of the first ground conductor layer 20b includes an inner point IP3 located below the outer point OP3 and inside the outer point OP3 in a portion where the resist layer 17a is in contact with the side surface in the inside-outside direction. The resist layer 17a is located under the portion P3 between the outer point OP3 and the inner point IP3 on the side surface of the first ground conductor layer 20b. Therefore, the resist layer 17a is caught by the portion P3. Thus, the resist layer 17a is prevented from being peeled off from the resin layer 16a. In this case, the first ground conductor layer 20b corresponds to the metal foil layer, the resist layer 17a corresponds to the upper resin layer, and the resin layer 16a corresponds to the lower resin layer.

(69) Further, when the high-frequency signal transmission line 10c is viewed in the downward direction, the second ground conductor layer 22b corresponds to the metal foil layer, the resist layer 17b corresponds to the upper resin layer, and the resin layer 16b corresponds to the lower resin layer. Therefore, the resist layer 17b is prevented from being peeled off from the resin layer 16b.

(70) As described above, the high-frequency signal transmission line 10c includes a plurality of sets of the upper resin layer, the lower resin layer, and the metal foil layer. More specifically, the high-frequency signal transmission line 10c includes three sets each having the upper resin layer, the lower resin layer, and the metal foil layer.

(71) Fourth Modification

(72) Hereinafter, a high-frequency signal transmission line 10d according to a fourth modification of a preferred embodiment of the present invention will be described with reference to the drawings. FIG. 12 is a cross-sectional view of the high-frequency signal transmission line 10d taken along line A-A.

(73) As illustrated in FIG. 12, the high-frequency signal transmission line 10d is different from the high-frequency signal transmission line 10 in that a signal conductor layer 18d having a shape different from that of the signal conductor layer 18 is provided. Note that since other configurations of the high-frequency signal transmission line 10d are the same as those of the high-frequency signal transmission line 10, descriptions thereof will be omitted.

(74) As illustrated in FIG. 12, the shape of the side surface of the signal conductor layer 18d is different from the shape of the side surface of the signal conductor layer 18. To be specific, a portion located below an inner point IP4 in the side surface of the signal conductor layer 18d is parallel or substantially parallel to the multilayer body up-down direction. Also in this case, similar to the high-frequency signal transmission line 10, the resin layer 16a is located below a portion P4 (see FIG. 12). Therefore, in the high-frequency signal transmission line 10d, the resin layer 16a is prevented from being peeled off from the resin layer 16b for the same reason as that of the high-frequency signal transmission line 10. In other words, when the resin layer 16a is located below the portion P4, the resin layer 16a is prevented from being peeled off from the resin layer 16b regardless of the shape of the side surface of the signal conductor layer 18d located below the inner point IP4.

(75) Other Preferred Embodiments

(76) It is not limited to the high-frequency signal transmission lines **10** and **10a** to **10d** according to the above-described preferred embodiments of the present invention, and it can be changed within the scope of the gist thereof. Further, it is possible to combine the configurations of the high-frequency signal transmission lines **10** and **10a** to **10d**.

(77) In the high-frequency signal transmission lines **10** and **10a** to **10d**, the resin layer **16a** and the resin layer **16b** are not necessarily made of the same material, and may be made of different materials.

(78) In the high-frequency signal transmission lines **10**, **10b**, **10b2**, **10c**, and **10d**, the outer points OP, OP2, OP3, and OP4 do not necessarily need to be located above the center in the multilayer body up-down direction on the side surfaces of the signal conductor layers **18** and **18b** and the first ground conductor layer **20b**. In addition, in the high-frequency signal transmission lines **10**, **10b**, **10b2**, **10c**, and **10d**, the inner points IP, IP2, IP3, and IP4 do not necessarily need to be located below the center in the multilayer body up-down direction on the side surfaces of the signal conductor layers **18a** and **18b** and the first ground conductor layer **20b**. In the high-frequency signal transmission lines **10**, **10b**, **10c**, and **10d**, and the side surface of the first ground conductor layer **20b**, the outer points OP, OP2, OP3, and OP4 are located above the center of the side surfaces of the signal conductor layers **18a** and **18b** and the first ground conductor layer **20b** in the multilayer body up-down direction, and the inner points IP, IP2, IP3, and IP4 may be located below the center of the side surfaces in the multilayer body up-down direction.

(79) In the high-frequency signal transmission lines **10** and **10a** to **10d**, the resin layers **16a** and **16b** are not necessarily be made of thermoplastic resin.

(80) In the high-frequency signal transmission lines **10** and **10a** to **10d**, the signal conductor layer **18** may be fixed to the resin layer **16a**.

(81) In the high-frequency signal transmission lines **10** and **10a** to **10d**, all the side surfaces of the signal conductor layer **18**, **18a**, **18b**, and **18d** may include the outer points OP and OP1 to OP4, and the inner points IP and IP1 to IP4. In this case, in all cross sections of the signal conductor layer **18** parallel or substantially parallel to the multilayer body up-down direction, the outer points OP and OP1 to OP4 are located above the inner points IP and IP1 to IP4 and the layer interface RI.

Alternatively, in the high-frequency signal transmission lines **10** and **10a** to **10d**, a partial side surface of the signal conductor layer **18** may include the outer points OP and OP1 to OP4, and the inner points IP and IP1 to IP4.

(82) The left side surface of the signal conductor layer **18** may have a left-right asymmetric structure with respect to the right side surface of the signal conductor layer **18**. Similarly, the left side surface of the signal conductor layer **18d** may have a left-right asymmetric structure with respect to the right side surface of the signal conductor layer **18d**.

(83) The left side surface of the first ground conductor layer **20b** may have a left-right asymmetric structure with respect to the right side surface of the first ground conductor layer **20b**. Similarly, the left side surface of the second ground conductor layer **22b** may have a left-right asymmetric structure with respect to the right side surface of the second ground conductor layer **22b**.

(84) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A resin multilayer substrate comprising: a metal foil layer, an upper resin layer, and a lower resin layer; wherein the metal foil layer includes a metal foil layer upper main surface, a metal foil layer lower main surface with a surface roughness greater than a surface roughness of the metal foil layer

upper main surface, and a metal foil layer side surface that is a surface of the metal foil layer excluding the metal foil layer upper main surface and the metal foil layer lower main surface; the metal foil layer upper main surface is in contact with the upper resin layer; the metal foil layer lower main surface is in contact with the lower resin layer; a portion of a lower main surface of the upper resin layer and a portion of an upper main surface of the lower resin layer are in contact with each other to define a layer interface; in a cross section of the metal foil layer parallel or substantially parallel to a multilayer body up-down direction, a direction from an inner side of a region surrounded by a surface of the metal foil layer toward an outer side portion of a region surrounded by the surface of the metal foil layer of directions perpendicular or substantially perpendicular to the multilayer body up-down direction is defined as an inside-outside direction; in a cross section of the metal foil layer parallel or substantially parallel to a multilayer body up-down direction, the metal foil layer side surface includes an outer point and an inner point located at a portion where the upper resin layer is in contact with the side surface; and the inner point is located below the outer point and is located inward of the outer point in the inside-outside direction.

2. The resin multilayer substrate according to claim 1, wherein a circuit pattern is defined by the metal foil layer.

3. The resin multilayer substrate according to claim 1, wherein the resin multilayer substrate has flexibility.

4. The resin multilayer substrate according to claim 1, wherein a signal conductor layer is defined by the metal foil layer.

5. The resin multilayer substrate according to claim 1, wherein the upper resin layer and the lower resin layer are made of a thermoplastic resin.

6. The resin multilayer substrate according to claim 1, wherein a metal foil layer is not fixed to the upper resin layer.

7. The resin multilayer substrate according to claim 1, wherein a material of the upper resin layer and a material of the lower resin layer are different from each other.

8. The resin multilayer substrate according to claim 1, wherein the upper resin layer is a protective film of the metal foil layer.

9. The resin multilayer substrate according to claim 1, wherein a plurality of sets each including the upper resin layer, the lower resin layer, and the metal foil layer is provided.

10. The resin multilayer substrate according to claim 1, wherein a close-contact strength of the metal foil layer to the lower resin layer is stronger than a close-contact strength of the metal foil layer to the upper resin layer.

11. The resin multilayer substrate according to claim 1, wherein the outer point is located above the inner point and the layer interface in all cross sections parallel or substantially parallel to the multilayer body up-down direction.

12. The resin multilayer substrate according to claim 1, wherein the outer point is located above a center of the side surface in the multilayer body up-down direction; and the inner point is located below a center of the side surface in the multilayer body up-down direction.
